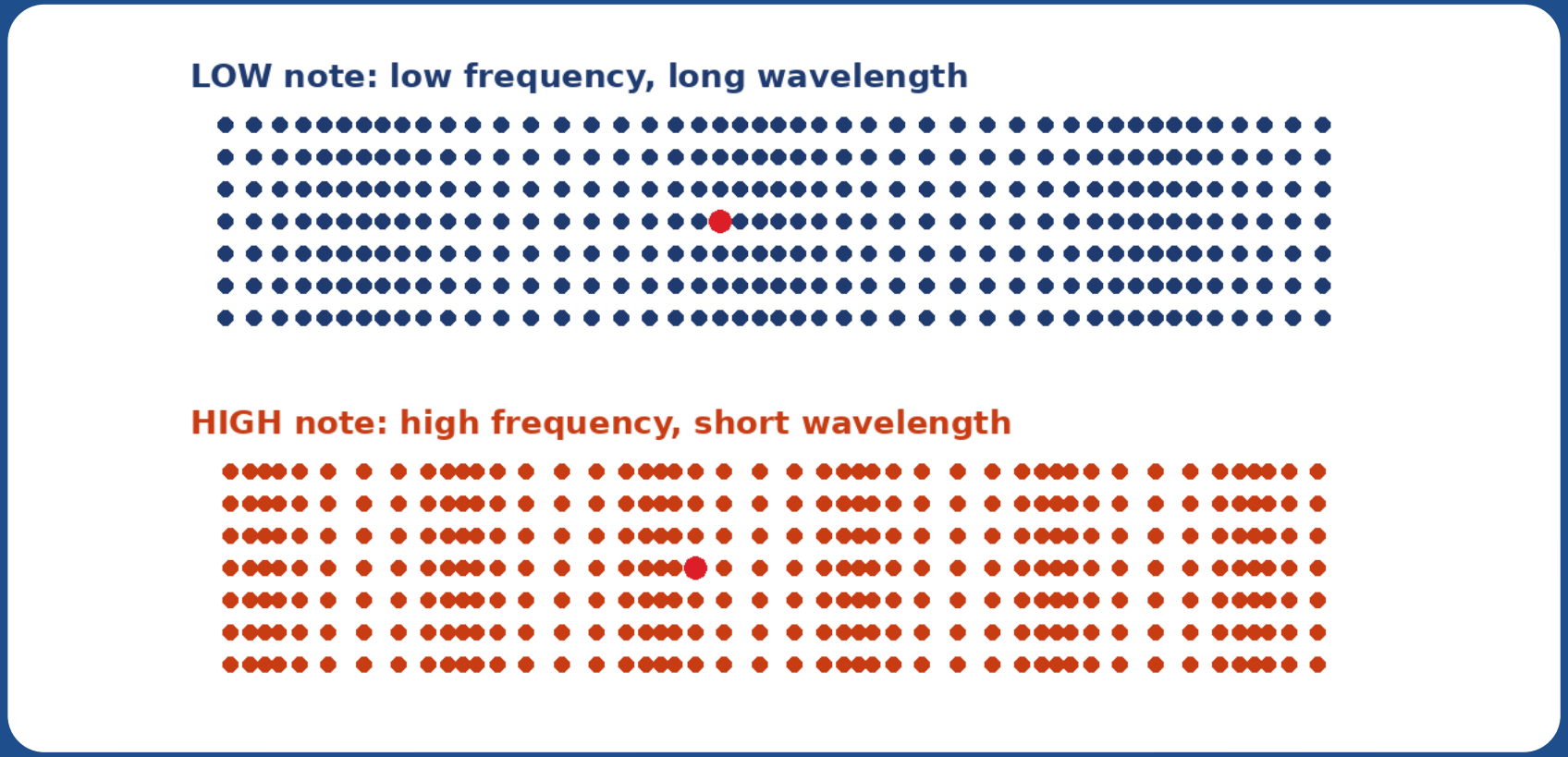
4.11 Sound Waves Lab

How does sound travel, and what changes pitch and loudness?



Think About It

You pluck a guitar string harder. What changes about the sound?

Single choice poll

A. It gets higher

Single choice poll

B. It gets louder

Single choice poll

C. It travels faster

Single choice poll

D. Its wavelength gets shorter

Today's Goals

I can explain how sound travels as a longitudinal wave.

I can connect pitch to frequency and wavelength.

I can connect loudness to amplitude and energy.

I can explain why sound needs a medium.

Rate each goal:

1 Not yet

2 Getting there

3 I can teach it

Ohio Science Standards

7.PS.4 Energy can be transferred through a variety of ways.

- Mechanical waves need a medium. The medium moves, then returns to its rest position.
- The energy of a mechanical wave increases with amplitude.
- Wave speed depends on the material: sound travels faster through most solids than gases.

In this lesson

- Integers: Case #2
- Part 1: Music Lab Sound Waves
- Pitch vs. loudness
- Part 2: PhET Sound Waves (4 screens)
- Can sound travel with no air?

Source: Ohio's Learning Standards for Science (2018), Grade 7 Physical Science; content statement paraphrased.

CASE #2 • TUESDAY

Was the sound quieter 5 seconds ago?

WAVE WORD: *Decibel*

The unit for measuring how loud a sound is (dB).

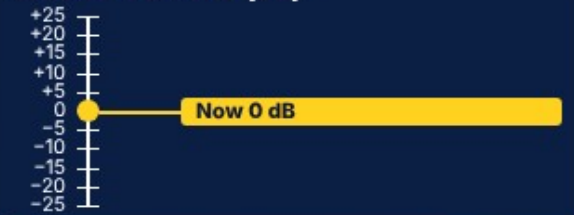
MYSTERY

In the sound lab, a student walks away from a speaker. The sound level changes by −4 decibels every second as she moves away. Her partner says that 5 seconds ago (−5 s), the sound was 20 decibels quieter than it is now. Is the partner right?

CLUES

- −4 dB per second
- Time: −5 s (the past)
- Partner says: quieter

SOUND VS. NOW (dB)





CLAIM

Is the partner right?

☐ YES ☐ NO

EVIDENCE

Solve:

(−4) × (−5) = _____

5 seconds ago, the sound was _____ dB _____.

REASONING

Finish the sentence:

The signs are _____, so the answer is _____. 5 seconds ago the sound was _____ dB _____ than now.

Today's Lab Plan

Use Chrome

Parts of the lab only work in the Chrome browser.

Part 1: Music Lab

Play low and high notes. Watch the dots.

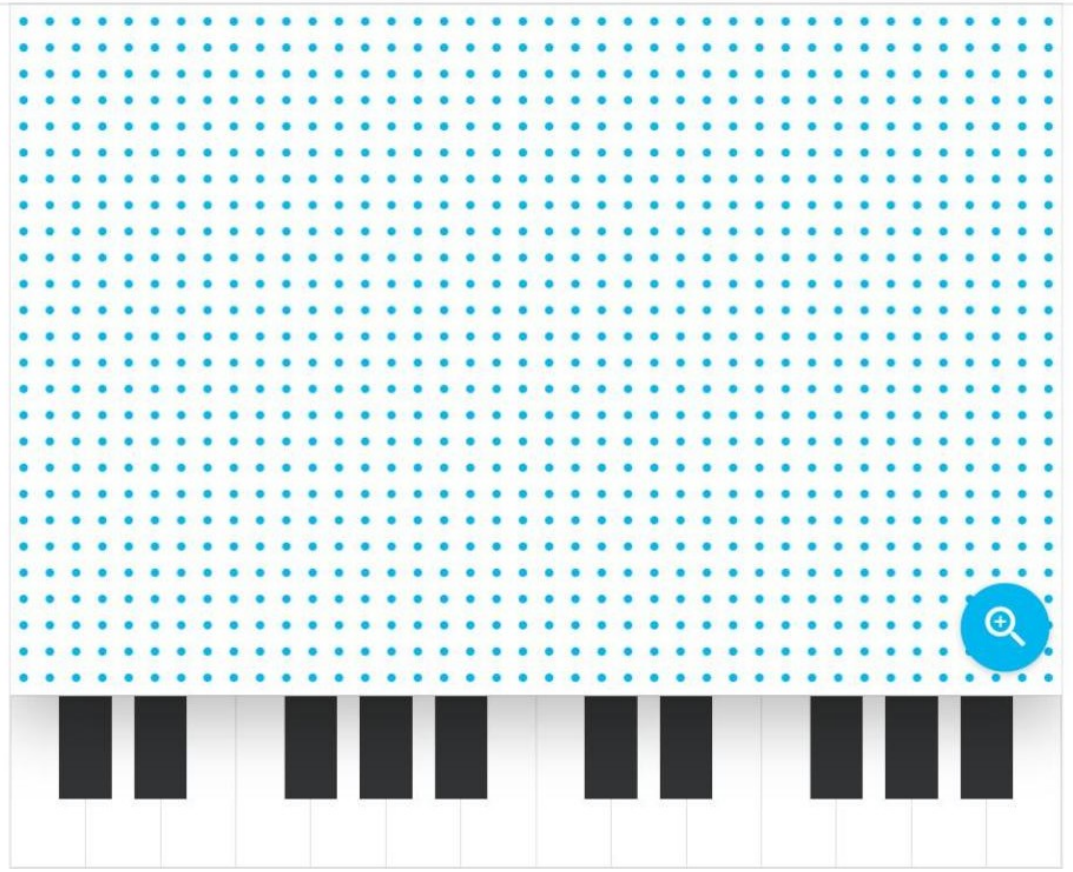
Q1–Q4 · about 15 min

Part 2: PhET Sound Waves

Intro, Measure, Two Sources, and Air Pressure screens.

Q5–Q9 · rest of class

Part 1: Music Lab Sound Waves



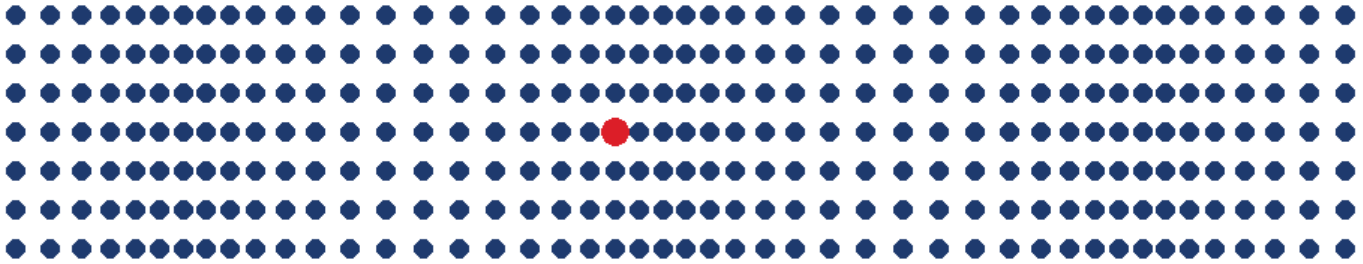
▶️ Open Music Lab: Sound Waves

1. Play a LOW note, then a HIGH note. Compare the dots. (Q1)
2. Watch the red particle. Where does it end up? (Q2)
3. Watch the blue dots. Which way do they move compared to the sound wave? (Q3)
4. A sound gets higher. What happens to the compressions? (Q4)

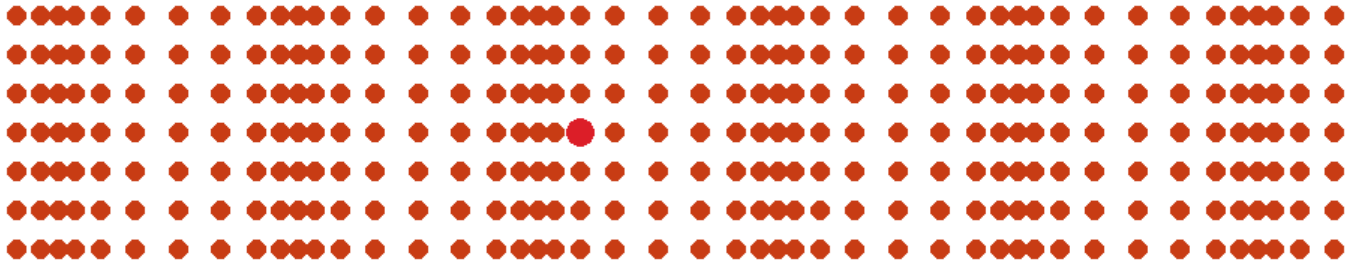
Classkick: 4.11 Part 1 · Q1–Q4

Low Note vs. High Note

LOW note: low frequency, long wavelength



HIGH note: high frequency, short wavelength



Same air, same speed: both waves travel equally fast.

Watch the red dot. It only moves back and forth. The energy moves on.

High note: particles vibrate more often, so the compressions are closer and the wavelength is shorter.

Engageli Poll

You turn UP the volume on a speaker. Which part of the sound wave changes?

Single choice poll

A. Frequency

Single choice poll

B. Amplitude

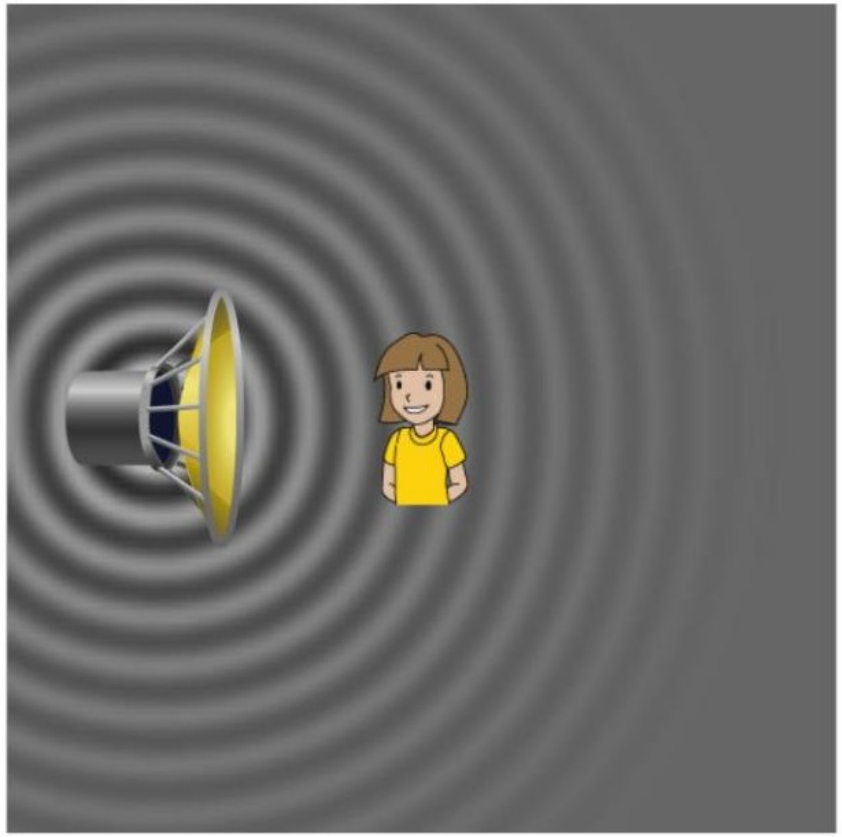
Single choice poll

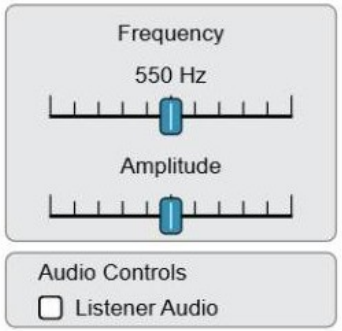
C. Wave speed

Single choice poll

D. Wavelength

Part 2: PhET Intro Screen





1. Check Listener Audio.
2. Slide the FREQUENCY up. Watch the rings and listen. (Q5)
3. Now slide the AMPLITUDE up. (Q6)

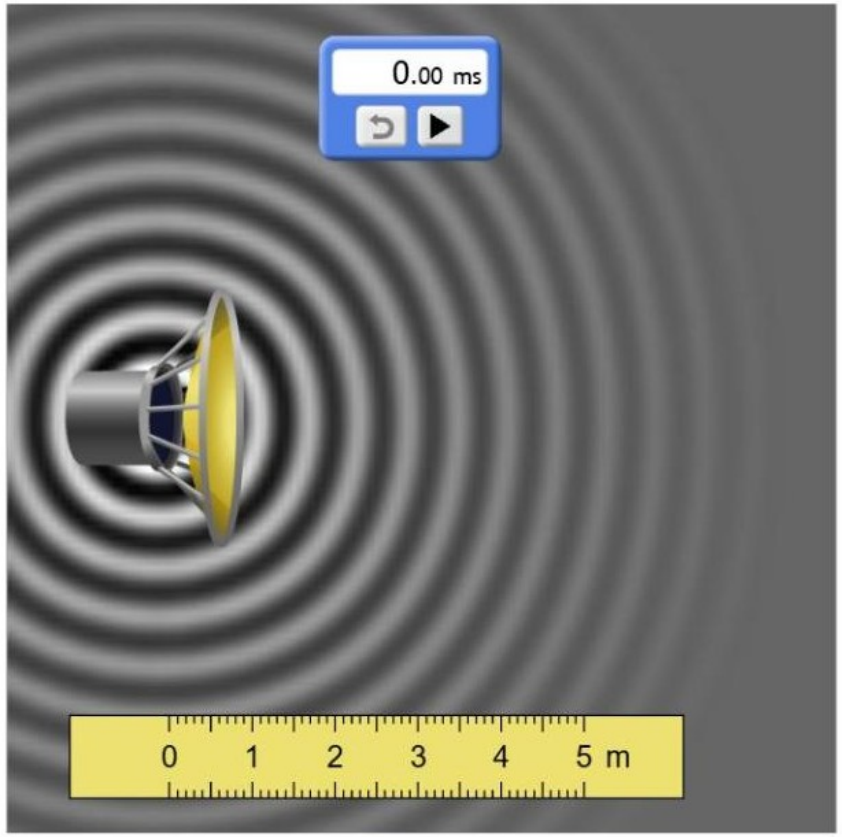
Open PhET Sound Waves

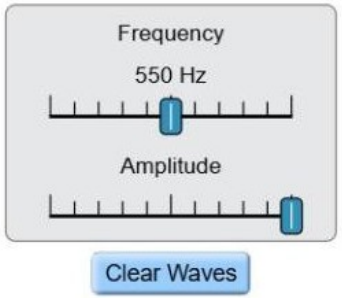
Compare: Pitch vs. Loudness

	Pitch (high or low)	Loudness (loud or soft)
Set by…	frequency	amplitude
Higher pitch / louder means…	higher frequency	bigger amplitude
Wavelength	gets shorter	stays the same
Energy	—	more energy
Wave speed in air	stays the same	stays the same

In PhET: rings closer together = higher pitch. Darker, brighter rings = louder.

Part 2: Measure Screen



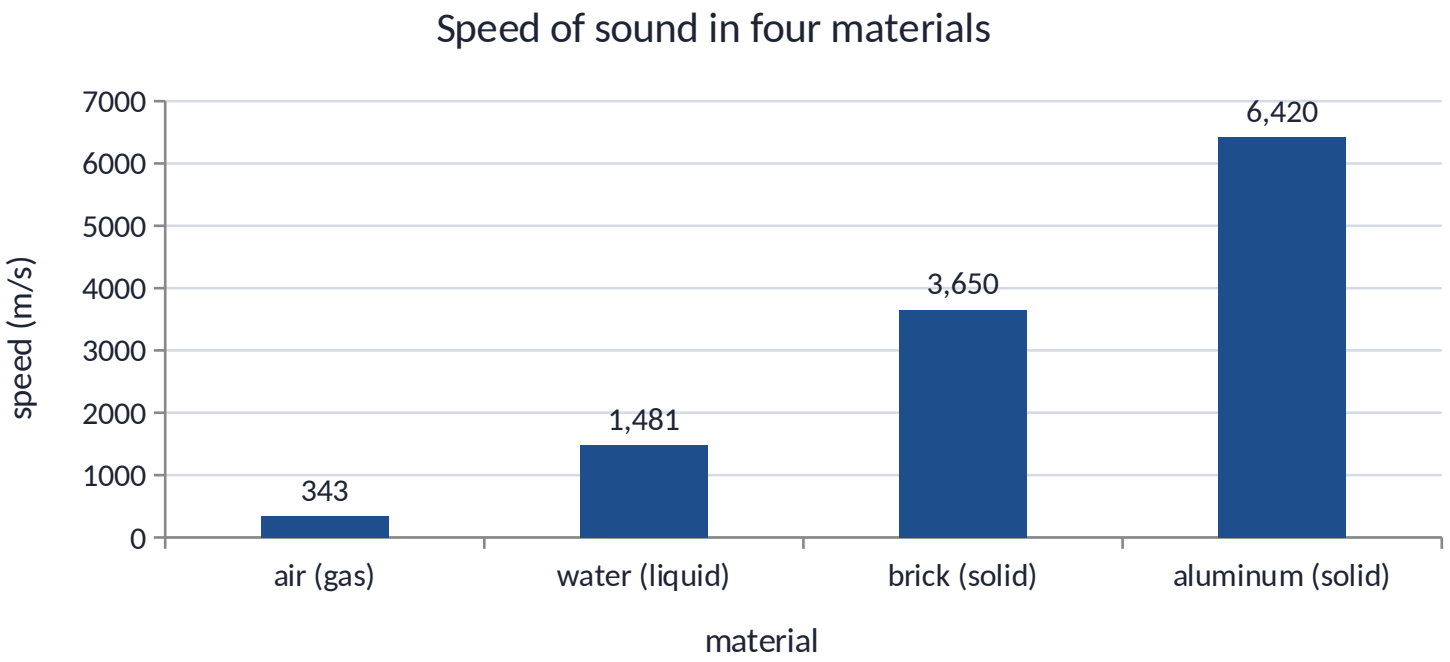


1. Set the frequency. Press Pause.
2. Drag the ruler from one light ring to the next light ring.
3. Record the wavelength in meters.

Frequency	Wavelength (m)
200 Hz	___
500 Hz	___
1000 Hz	___

Open PhET Sound Waves

Graph Poll



Read the graph.

What pattern does the graph show?

Single choice poll

A. Fastest in gases

Single choice poll

B. Fastest in solids

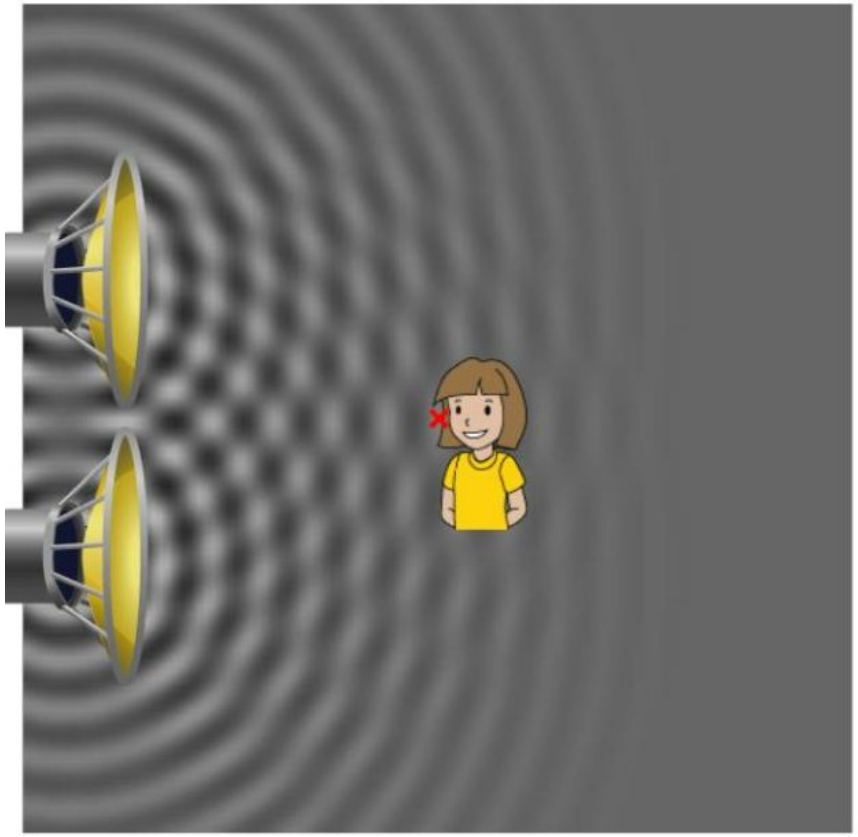
Single choice poll

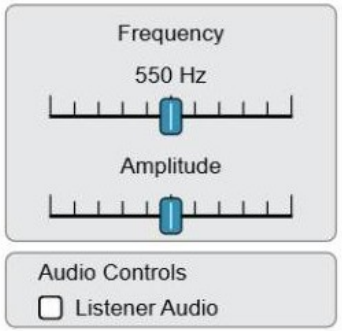
C. Same in all

Single choice poll

D. Fastest in liquids

Part 2: Two Sources Screen

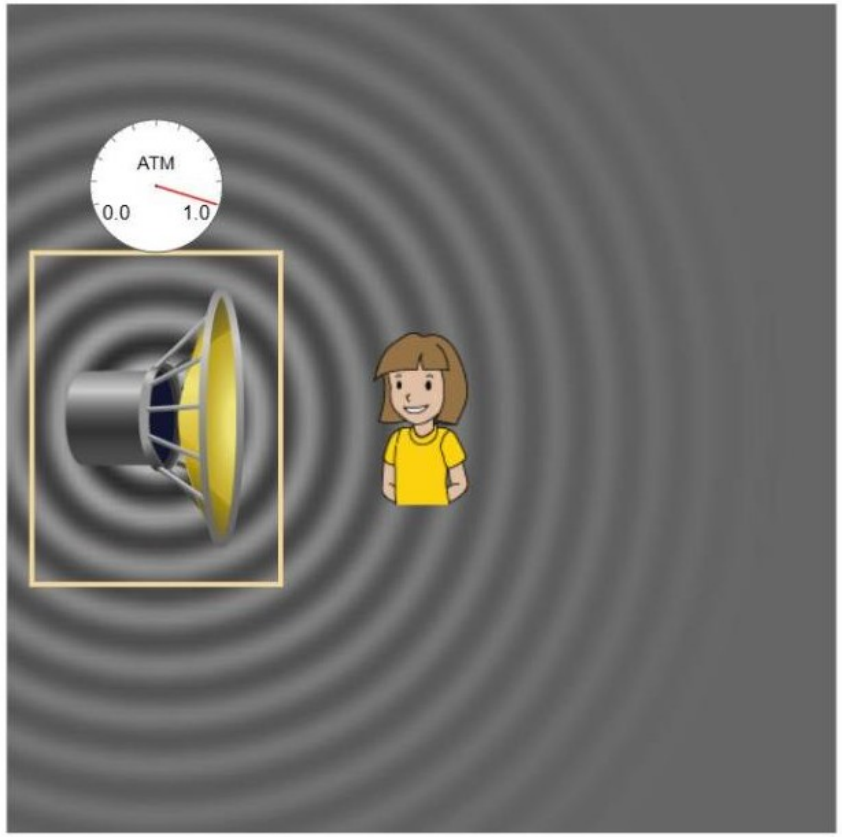


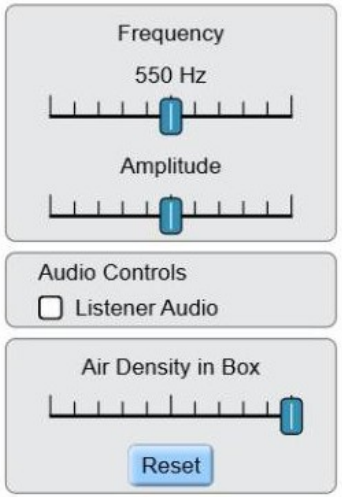


1. Check Listener Audio.
2. Drag the girl up and down. Listen for loud and quiet spots.
3. Loud spot: crest meets ___. Quiet spot: crest meets ___. (Q8)

▶ Open PhET Sound Waves

Part 2: Air Pressure Screen





1. Check Listener Audio.
2. Slide Air Density in Box all the way down.
3. Watch the waves leaving the box. Listen. (Q9)

Open PhET Sound Waves

Who Is Right?

In a space movie, a ship explodes and you hear a huge BOOM. Could that really happen in space?

Ava says:

"Yes. Explosions are so loud that the sound can reach anywhere."

Ben says:

"No. Sound needs a medium, and space has almost no air to carry it."

Cam says:

"Yes. The light from the explosion carries the sound with it."

Dee says:

"Yes, but you hear it later because space is so big."

Single choice poll

A. Ava

Single choice poll

B. Ben

Single choice poll

C. Cam

Single choice poll

D. Dee

Key Ideas

1. Sound is a longitudinal, mechanical wave made by vibrations.
2. Higher pitch = higher frequency = shorter wavelength.
3. Louder = bigger amplitude = more energy. The wavelength stays the same.
4. Sound needs a medium. No air = no sound. It is fastest in solids.

Re-rate your goals. Not done? Finish Part 2 on your own.